

ENGR 181W – Spring 2026

Technical Communication and Documentation

Instructor: Tofigh Heidarzadeh, PhD

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Please do not forget to add **Course Name / number** and **Workshop Sec. number** in the Subject line of your emails.

Lectures:

MWF, 4:00 – 4:50 PM: Student Success Center (SSC), Room 329

Labs:

LAB 021 (Monday): 1:00-3:50 PM, Sproul Hall 2355

LAB 022 (Wednesday): 1:00-3:50 PM, Watkins Hall, Room 1111

Office Hours: Friday, 11:30 AM- 12:30 PM

Office Location: 307 Winston Chung Hall

Labs:

LAB 021 (Monday): 1:00-3:50 PM, Sproul Hall, Room 2355

LAB 022 (Wednesday): 1:00-3:50 PM, Watkins Hall, Room 1111

WAC writing TA: Jovana Iveski, Email: jisev001@ucr.edu

WAC writing tutor (undergraduate writing consultant): Carissa Le, Email: carissa.le@email.ucr.edu

This course develops technical communication and documentation skills. Includes writing and critiquing design documentation and technical reports. Uses the capstone design project in an engineering discipline as a case study. Considers professional and ethical responsibilities, contemporary issues in engineering and global impact on economics, society, and the environment.

Fulfills the third- quarter writing requirement for students who earn a grade of "C" or better for courses that the Academic Senate designates, and that the student's college permits, as alternatives to English 001C.

Prerequisites:

ENGL 001B with a grade of "C" or better; Senior standing and completion of one quarter of a senior design project in an engineering discipline.

Students also need to register for ENGL 007-014 in the first two weeks of the quarter (if have not done so already.) The CRN for ENGL 007-014 is 66764. ENGL 007-014 has been titled “ENGR181W Writing Workshop” on Banner.

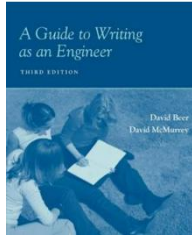
Course Topics:

This course is tightly integrated with the senior design project course in engineering disciplines so that students can use first-hand knowledge of their actual design projects in their writing and analysis. Students are required to write about a wide range of engineering specific topics, such as documentation of design specifications and processes, description of systems, structures, hardware and/or software, test plan, test report, user interface, product marketing, industrial standards, realistic constraints, ethical and environmental impact of their designs.

Requirements:

- Attendance & participation
- Satisfactory completion of all required assignments

Required Texts and Equipment:



- D. F. Beer and D. A. McMurrey, *A Guide to Writing as an Engineer*, 4th ed. New York: Wiley Publications, 2014.

Note: A DPF copy of this book will be posted on Canvas (WEEK 1 module).

- W. Pfeiffer, *Technical Communication: A Practical Approach*, 8th ed. Vancouver, B.C: Langar College, 2014.
- **Note:** I will use the 8th edition of this book, however, the 7th edition is also usable.
- IEEE Citation Reference,
<https://www.ieee.org/documents/ieeecitationref.pdf>



Recommended Sources:

- L. Williamson, *On demand writing; applying the strategies of impromptu speaking to impromptu writing. The Free Library* (May 1), [https://www.thefreelibrary.com/On demand writing; applying the strategies of impromptu speaking to...-a0199021069](https://www.thefreelibrary.com/On-demand-writing;applying-the-strategies-of-impromptu-speaking-to...-a0199021069) (accessed March 11 2022)
- N. Ramsey, "Learning Technical Writing, Using the Engineering Method, A Handbook for Groups," <https://www.cs.tufts.edu/~nr/pubs/learn.pdf>
- G. J. Alfred, et al, *Handbook of Technical Writing*, 9th ed. Boston: Bedford/St. Martin's, 2009. An online PDF copy of this book is accessible at:
<http://site.iugaza.edu.ps/mmushtaha/files/Handbook-of-Technical-Writing-9th-Edition.pdf>
- D. J. Weatherford, *Technical Writing for Engineering Professionals*, New York: PennWell Books, 2016.

Useful Online Sources:

- <http://tc.eserver.org/>
- <https://www.prismnet.com/~hcexres/textbook/>
- <https://ewh.ieee.org/soc/es/Aug1996/030/cd/write/begin.htm>
- <http://www.seas.upenn.edu/faculty/tech-comm/>
- [http://learningpath.org/articles/Free Online Technical Writing Courses from Top Universities.html](http://learningpath.org/articles/Free-Online-Technical-Writing-Courses-from-Top-Universities.html)

Class Policies:

1. Cheating, plagiarism, disruptions of instructional activity, fraud and/or lying will result in, at a minimum, a grade of "F" for the course, as well as formal disciplinary action based on Academic Integrity Policies & Procedures.
2. All assignments must be turned into Canvas. You have to run each assignment through Turnitin during the assigned time interval and turn in a hard copy to the instructor.

3. All assignments are due at the beginning of class on the day that they are due. Late assignments will not be accepted.
4. All assignments must follow the exact format and rules which will be announced during the first week of instruction.
5. All instructions, rules, and assignments will be explained in lecture and lab. It is the students' responsibility to attend lectures and lab for this information.

Weekly Lectures:

Week 1: March 30- April 3 - The writing process, writing a good introduction.

Readings: Beer, Ch. 1

Lab: Principles of technical writing. How to write an Executive Summary, and Writing an Introduction of your design project that includes Design Objectives and System Overview, Backgrounds and Prior Art, Development Environment and Tools (>750 words)

Week 2: April 6 – April 10 - Logic and persuasion, inductive and deductive reasoning, outlining, mind maps.

Readings: Beer, Ch. 2

Lab: Ways to think and talk about engineering design. Writing an introduction of your design project that includes Design Objectives and System Overview, Backgrounds and Prior Art, Development Environment and Tools (>750 words)

An analysis of design considerations includes Assumptions, Realistic Constraints, Industry Standards, and Budget and Cost Analysis (>750 words)

Week 3: April 13 – April 17 - Grammar and style of technical writing

Lab: Eliminating noise in technical writing. Useful rules of grammar, punctuation, abbreviation. An analysis of design considerations includes Assumptions, Realistic Constraints, Industry Standards, and Budget and Cost Analysis (>750 words).

A report on Experiment Design and Feasibility Study that designs experiments and data to evaluate competing design and implementation. (>750 words)

Week 4: April 20 – April 24 - Presentation of data and analysis, use of graphs and tables, visual gestalt and page layout

Lab: Using information wisely. A report on Experiment Design and Feasibility Study that designs experiments and data to evaluate competing design and implementation. (>750 words). A technical report documenting the hardware and software design, including architecture, high level design, block diagrams, flowcharts and user interface (>2000 words)

Week 5: April 27 – May 1 - Writing functional specifications, design document and test report +
Practice: 'How to write a Progress Report'

Lab: Using information wisely. A technical report documenting the hardware and software design, including architecture, high level design, block diagrams, flowcharts and user interface (>2000 words).

A test plan and test report that describes the procedures for the tests performed, presents the test results using charts and graphs, and provides an analysis of the test data relative to the design specification (>1000 words)

Week 6: May 4 – May 8 - Use cases/scenarios, persona and scenario writing

Lab: A test plan and test report that describes the procedures for the tests performed, presents the test results using charts and graphs, and provides an analysis of the test data relative to the design specification (>1000 words)

- **Writing an essay on “Understanding of Professional and Ethical Responsibility and Global, Economic, Environmental and Societal Impact of your project” (>1300 words).**

Note: Essays are not a group activity. Each team member must write their essays independently. To avoid overlap, team members may choose to address different sub-questions of the prompt.

- a) **In this essay you have to explain:** What are the ethical implications of your project if your design becomes a commercial product or service, examples include but not limited to implications in health, education, privacy, social attitude, fairness of access, privacy, public safety, psychological and developmental effects on children and young adults, etc.?
- b) How does your approach address these ethical issues?
- c) What you learned through this design project about professional and ethical responsibility.
- d) Discuss the potential global, economic, societal, and environmental impact of the project if your design becomes a commercial product or service. You do not need to address every aspect, just focus on a couple of aspects that are related to your project. For example, if your design becomes a commercial product or service, how will it improve quality of life, affect the environment, energy usage, resource allocation, enhance entertainment, education, globalization etc.? Are there any political debates, laws and regulations that are related to your project?

Week 7: May 11 – May 15 - Analyzing ethical responsibilities, contemporary issues and impacts + Practice: ‘How to write a Proposal Letter’

Lab: Professional expectations of ethics and integrity.

- A technical report documenting the hardware and software design, including architecture, high level design, block diagrams, flowcharts and user interface (>2000 words)

Week 8: May 18 – May 22 – Resume and Interview + Project Presentations

Lab: Conclusions and related outcomes: Write a conclusion of the project and discuss future improvements (>400 words)

- **Write an essay about “Contemporary Engineering Issues and Recognition of the need for and an ability to engage in lifelong learning” (>1000 words). In this essay you have to:**

- a) Discuss the contemporary engineering issues related to the project. Potential contemporary engineering issues related to your project include new technologies, new industry standards, new design methods, new materials, new trends in manufacturing, etc. You do not need to address every aspect, just focus on a couple

of aspects that are related to your project. Example: One trend in electronics is high level integration and very low power consumption requirement, how is that related to your project? Another example is Internet-connected distributed real-time sensing, how does your project fit with that trend? Other examples of contemporary engineering issues are cloud computing, broadband wireless data becoming more widespread, renewable energy, natural resources becoming increasingly limited, population becoming old and needing more care, Internet security threats are increasing, etc.

- b) Discuss how doing this design project helped you (a) recognize the need, and (b) develop the ability in lifelong learning.

Week 9: May 27 – May 29 (May 25 is a holiday) – PPT Presentation

Lab: Executive summary.

- Write an executive summary that succinctly conveys the motivation and importance of their design, the market demand, and highlights the technical parameters, benefits and limitations of their design. It needs to capture the reader's attention and persuade the user of the importance of the design project. (Single page, 300<words<600)

Week 10: June 1 – June 5 - PPT Presentation

- Write an essay about “Project Management and Importance of Team Work” (>800 words). You have to:

- a) Explain how the project was managed. How were the tasks distributed? How was the schedule made? What project management software/method did you use? What have you learned in terms of teamwork, time management, project management etc., from this project?
- b) Discuss how the experience of working on the design project helped you understand the importance of teamwork. If your team includes members from more than one engineering discipline, describe what you learned from working in a multidisciplinary team. If your team is all from one engineering discipline, describe how you sought assistance or input from people in other disciplines (e.g., electrical engineering, mechanical engineering, computer science, bioengineering, physics, chemistry, etc.).

All writings will be evaluated by the instructor and/or TA. Feedback will be provided on presentation of ideas, reasoning, development, grammatical correctness, and clarity of assertions, logic of paragraphs, and the use of evidence. Students are required to revise their writing based on this feedback.

Grading (Percentage):

Writing Assignments: 70%

(Final edition of the report: 45%, Essays: 25%)

Presentations: 15%

Attendance of Workshop: 15%

Workshops (held by WAC):

Attendance of workshops is **required** and accounts for 15% of grades. Students in the workshops develop their writings, give presentations and receive critiques. Workshops aimed at developing specific writing skills will also be provided. The workshops will be supported by the campus WAC Program.

The workshops would focus on the use of writing and reading to address the specific requirements of the writing assignments set forth by the course instructors. The TA would offer enough workshop sections to accommodate all ENGR 181W students in groups of approximately fifteen. Thus, during the quarter, the students would attend three fifty-minute WAC workshops during their workshop periods. In addition, a tutor, selected and trained by the UWP, would be available to meet struggling students on an individual basis. The tutor would offer a minimum of ten appointment times per week, and as many as twenty if needed. Appointments would be made via a sign-up system on the ENGR 181W Canvas system.

Note: If two or more members of the same project are taking this course, each member's contribution in editing/correction of the report **MUST** be documented. In Word/docx, all contributions must be seen under Comments or Reviewing Pane and in Google Docs under Version History.

The final version of the report will not be graded without these documented inputs.

Week 1	Mon. March 30	Wed. April 1	Fri. April 3
	181W Lecture: 4:00-4:50 Lab 021- 1:00-3:50	181W Lecture: 4:00-4:50 Lab 022- 1:00-3:50	181W Lecture: 4:00-4:50
Week 2	Mon. April 6	Wed. April 8	Fri. April 10
	181W Lecture: 4:00-4:50 Lab 021- 1:00-3:50	181W Lecture: 4:00-4:50 Lab 022- 1:00-3:50	181W Lecture: 4:00-4:50
Week 3	Mon. April 13	Wed. April 15	Fri. April 17
	181W Lecture: 4:00-4:50 Lab 021- 1:00-3:50	181W Lecture: 4:00-4:50 Lab 022- 1:00-3:50	181W Lecture: 4:00-4:50
Week 4	Mon. April 20	Wed. April 22	Fri. April 24
	181W Lecture: 4:00-4:50 Lab 021- 1:00-3:50	181W Lecture: 4:00-4:50 Lab 022- 1:00-3:50	181W Lecture: 4:00-4:50
Week 5	Mon. April 27	Wed. April 29	Fri. May 1
	181W Lecture: 4:00-4:50 Lab 021- 1:00-3:50	181W Lecture: 4:00-4:50 Lab 022- 1:00-3:50	181W Lecture: 4:00-4:50
Week 6	Mon. May 4	Wed. May 6	Fri. May 8
	181W Lecture: 4:00-4:50 Lab 021- 1:00-3:50	181W Lecture: 4:00-4:50 Lab 022- 1:00-3:50	181W Lecture: 4:00-4:50 Essay 1 Due
Week 7	Mon. May 11	Wed. May 13	Fri. May 15
	181W Lecture: 4:00-4:50 Lab 021- 1:00-3:50	181W Lecture: 4:00-4:50 Lab 022- 1:00-3:50	181W Lecture: 4:00-4:50
Week 8	Mon. May 18	Wed. May 20	Fri. May 22
	181W Lecture: 4:00-4:50 Lab 021- 1:00-3:50	181W Lecture: 4:00-4:50 Lab 022- 1:00-3:50	181W Lecture: 4:00-4:50 Essay 2 Due
Week 9	Mon. May 25	Wed. May 27	Fri. May 29
	Holiday	181W Lecture: 4:00-4:50 Lab 022- 1:00-3:50	181W Lecture: 4:00-4:50
W	Mon. June 1	Wed. June 3	Fri. June 5

	181W Lecture: 4:00-4:50 Lab 021- 1:00-3:50	181W Lecture: 4:00-4:50 Lab 022- 1:00-3:50	181W Lecture: 4:00-4:50 Essay 3 Due
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